Yu-Hsuan Tsai

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EDUCATION

National Taiwan University Taipei, Taiwan
M.S. in Computer Science and Information Engineering Sep 2026 – Expected 2028
Web Mining and Information Retrieval Lab; Advisor: Prof. Pu-Jen Cheng.
Research interests: Information retrieval and machine learning.

National Yang Ming Chiao Tung University Hsinchu, Taiwan
B.S. in Computer Science Sep 2022 – Jun 2026
Relevant Coursework: Data Structures & OOP (A), Algorithms (A),
Artificial Intelligence (A+), Software Engineering (A+).

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, C#, Swift.
Software: React, Next.js, Node.js, Electron, PySide6, WebSocket, Git, Unity.
ML / Data: PyTorch, YOLO, OpenCV, NumPy, pandas; image and signal processing.

EXPERIENCE

Graduate Researcher Sep 2026 – Present
Multimodal Retrieval and Question Answering Project
- Ongoing research on dense and hybrid retrieval over Traditional Chinese text–image corpora.

Software Intern Mar 2026 – Sep 2026
Logitech, Mechatronics Team
- Developed a Python desktop tool for keyboard backlight evaluation, combining measurement processing, YOLO-based key detection, image matching, and automated test reporting.
- Built TypeScript/React interfaces and Node.js WebSocket services for peripheral configuration, calibration, and live telemetry; integrated device communication and debugged firmware/SDK issues.
- Built a Python desktop tool for input capture, gesture recognition, and configurable system actions.

Teaching Assistant, Systems Programming Sep 2026 – Present
National Taiwan University
- Design programming assignments and examination questions; develop automated correctness tests.

Research Assistant, Cognitive Engineering Lab 2024 – 2025
National Yang Ming Chiao Tung University
- Built Python pipelines for EEG preprocessing, feature extraction, and real-time sensor-data integration.

PROJECTS

OpenLogi 2026
Open-source contribution | Rust | 23k+ GitHub stars.
Local-first desktop application with an IPC-based client–agent architecture for Logitech device control.
- Strengthened configuration-state management by preserving short/long-press binding semantics; designed regression coverage for six button types, persistence, profile isolation, and failure rollback.

LiveSubtitle — iOS Subtitle and Translation App 2026
- Developed a Swift iOS app integrating native video playback, Japanese speech recognition, and on-device translation with time-aligned subtitle rendering.
- Implemented asynchronous audio processing, subtitle caching, seek cancellation, and local language-model inference through llama.cpp.

XR Interaction Projects — Unity / C# / Meta Quest 2025
- Built two interactive experiences with physical–virtual interaction and gesture controls; received Bronze and Special Presentation Awards at the NYCU XR Showcase.